

Course Equivalency Management System Using AI Assistance

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Project Idea:

- The proposed system automates the course equivalency process by comparing courses previously completed at external universities with those offered in the current university's study plan.
- The process operates at the course level, allowing students to upload course information and supporting documents, submit equivalency requests, and track decisions efficiently.
- The system incorporates AI-based similarity techniques to analyze course content semantically, providing intelligent recommendations that assist decision-makers and improve both accuracy and processing speed.

Problem Statement:

- Currently, course equivalency requests are handled manually, requiring considerable time and effort from academic staff.
- This process is often inefficient, as it involves repeated evaluation of similar courses and delays in decision-making.
- The proposed system addresses these challenges by automating the equivalency process and utilizing AI-based similarity matching techniques to reduce redundancy, enhance accuracy, and significantly improve decision speed.

Project Objectives:

- The primary objective of the proposed system is to automate and enhance the course equivalency process by integrating intelligent AI-based similarity analysis.
- The system specifically aims to:
 - - Automate the manual equivalency workflow to improve operational efficiency.
 - - Reduce redundancy in evaluating previously reviewed courses.
 - - Utilize AI techniques to analyze and compare course content effectively.
 - - Assist academic staff in making faster and more accurate decisions.
 - - Provide a user-friendly platform for students to submit and track requests.
 - - Maintain a structured database of past equivalency decisions for future reuse.

System Users:

- - Student:
Uploads transcripts, submits equivalency requests, and tracks request status.
- Doctor:
Reviews equivalency requests and makes decisions by approving or rejecting them.
- Administrator:
Manages universities, courses, and departments, and controls system users.

System Features:

- Upload and manage course transcripts in PDF format
- Submit course equivalency requests
- Track the status of submitted requests in real-time
- Perform AI-based semantic similarity analysis using the (text-embedding-3-small model)
- Provide a structured workflow for doctors to review and approve/reject requests
- Store and reuse past equivalency decisions to reduce redundancy

Functional Requirements:

- The system shall allow students to upload course transcripts in PDF format.
- - The system shall allow students to add course details (name, description, credit hours, university).
- - The system shall allow students to submit course equivalency requests.
- - The system shall allow students to track the status of their requests (Pending, Approved, Rejected).
- - The system shall extract text from uploaded PDF files.
- - The system shall perform AI-based similarity analysis between courses.

Functional Requirements:

- The system shall calculate and store similarity scores for each request.
- - The system shall allow doctors to view all pending equivalency requests.
- - The system shall allow doctors to review course details and uploaded documents.
- - The system shall allow doctors to approve or reject requests.
- - The system shall store decisions along with comments and timestamps.

Non-Functional Requirements

- - Performance:
The system should process similarity calculations within a reasonable time.
- Scalability:
The system should support a growing number of users and requests without performance degradation.
- Security:
The system should ensure secure authentication and protect user data.
- Usability:
The system should provide a simple and user-friendly interface for all users.

Non-Functional Requirements

- Reliability:

The system should operate continuously with minimal downtime.

- Maintainability:

The system should be easy to maintain, update, and extend.

- Availability:

The system should be accessible to users at all times.

- Accuracy:

The AI module should provide accurate similarity results to support decision-making.

System Architecture:

- Frontend: React.js
- Backend: [ASP.NET](#) Core Web API
- Database: SQL Server
- AI Module: text-embedding-3-small

System Workflow:

1. The student uploads a course transcript in PDF format
2. The system extracts and processes course information from the uploaded document.
3. The system checks existing equivalency records to identify similar past decisions.
4. The AI module performs similarity analysis and suggests potential equivalent courses.
5. The doctor reviews the request, including course details, uploaded files, and AI recommendations.
6. The final decision (approved or rejected) is stored in the system for future reference and reuse.

AI Role in the System:

- The AI module uses embedding-based similarity models to compare course content at a semantic level.
- It processes course descriptions and uploaded documents, converts them into numerical vectors,
- and calculates similarity scores to identify the most relevant equivalent courses.
- This significantly improves decision accuracy and assists doctors in evaluating requests efficiently.

Expected Benefits:

- Reduce the time required to process course equivalency requests
- Minimize the workload on doctors by automating repetitive tasks
- Improve the accuracy and consistency of equivalency decisions
- Enable reuse of previous decisions to avoid redundant evaluations
- Enhance the overall student experience through a faster and more transparent process

Activity Diagram:user

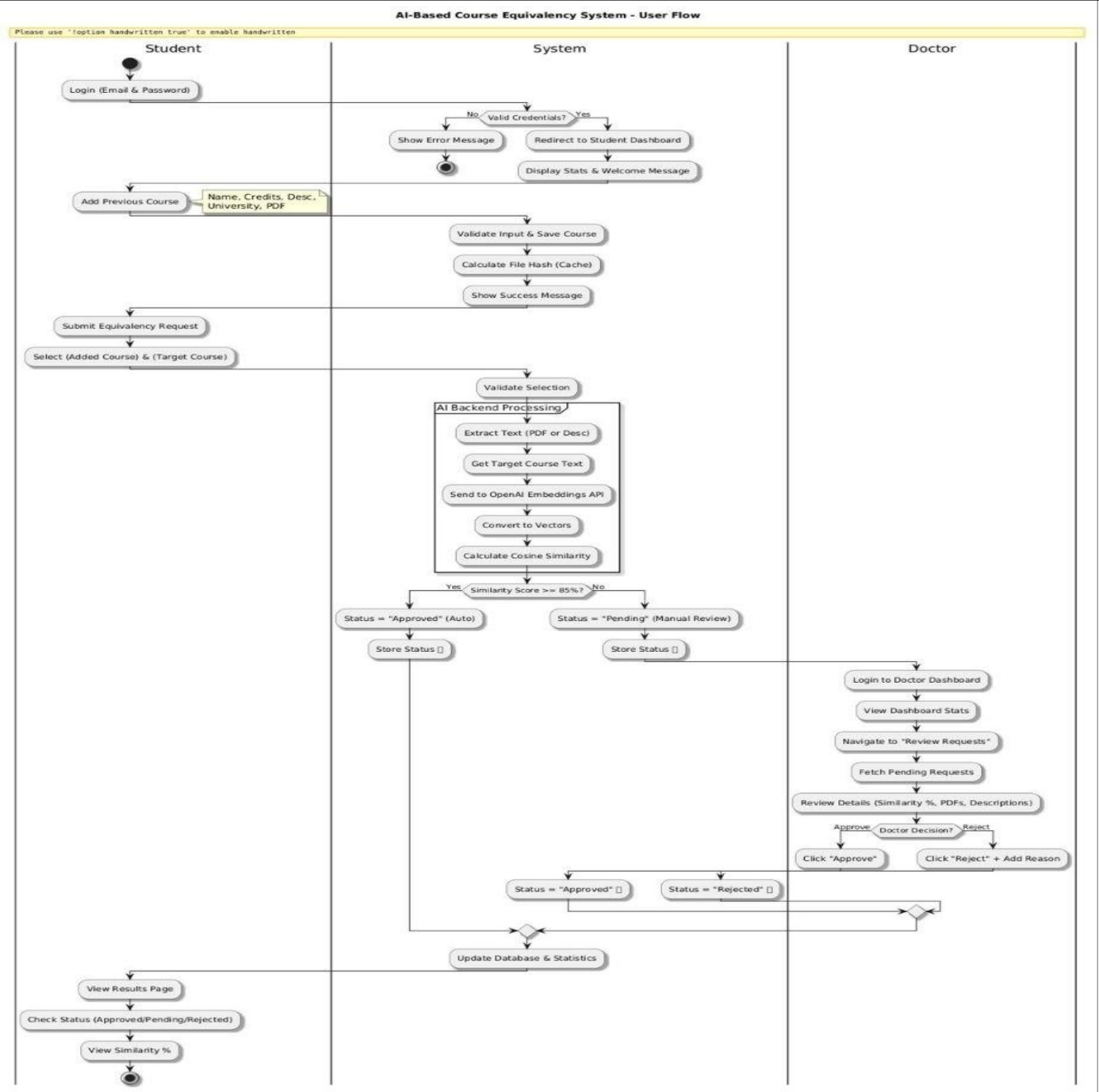
1:The system starts when the student uploads their course and submits an equivalency request.

2:The backend extracts the course content and uses an AI model to compute similarity between courses.

3:If the similarity is high, the system automatically approves the request.

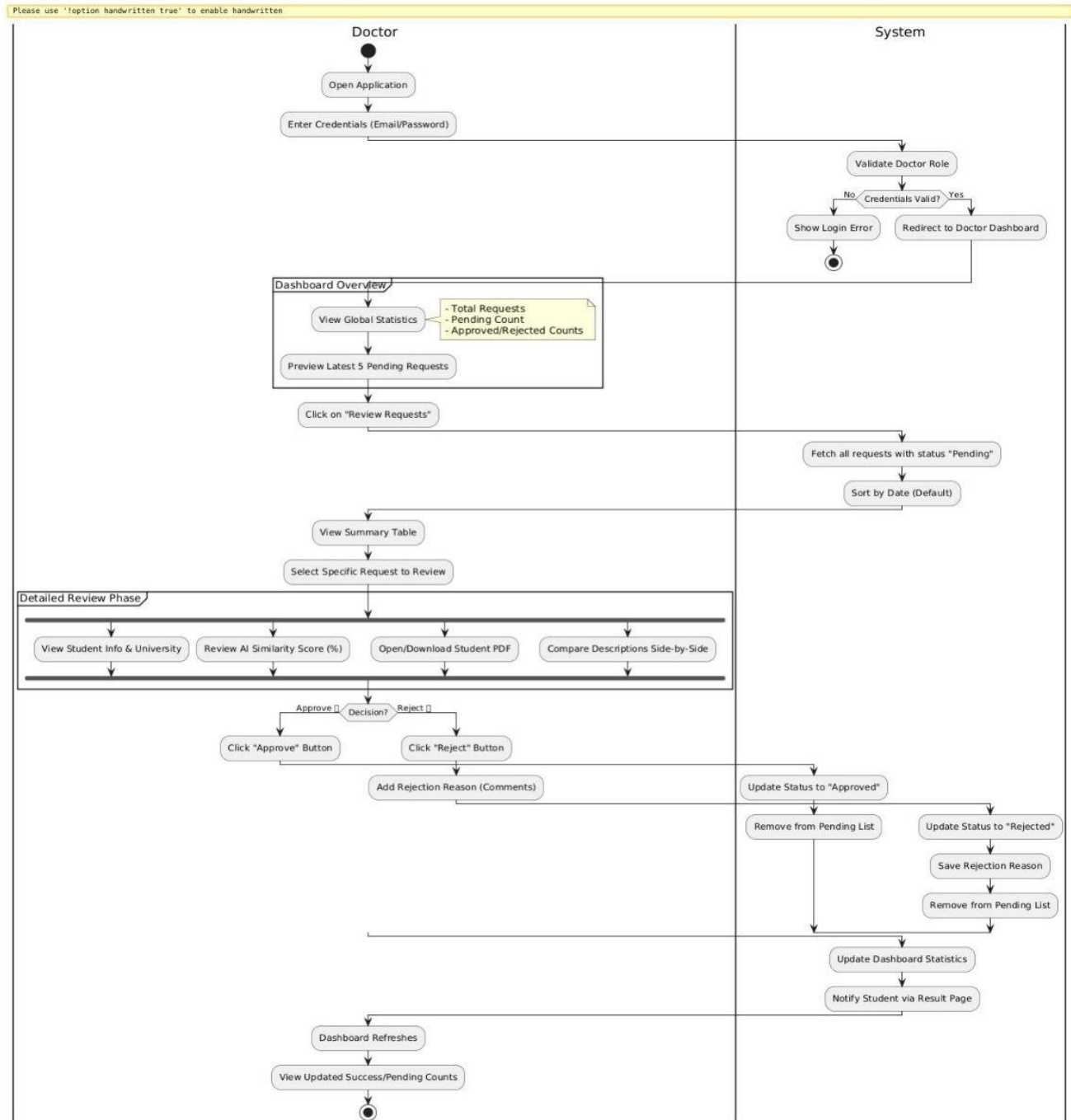
4:Otherwise, the request is sent to a doctor for manual review.

5:Finally, the decision is stored and displayed to the student.



Activity Diagram: Doctor

- 1: The doctor logs into the system and accesses the dashboard,
- 2: where they can view statistics and pending requests.
- 3: They select a request and review detailed information, including student data, AI similarity score, and course documents.
- 5: Based on this information, the doctor either approves or rejects the request.
- 6: The system then updates the request status, saves the decision, and notifies the student while updating dashboard statistics.



Admin Operations - Simplified Activity Diagram

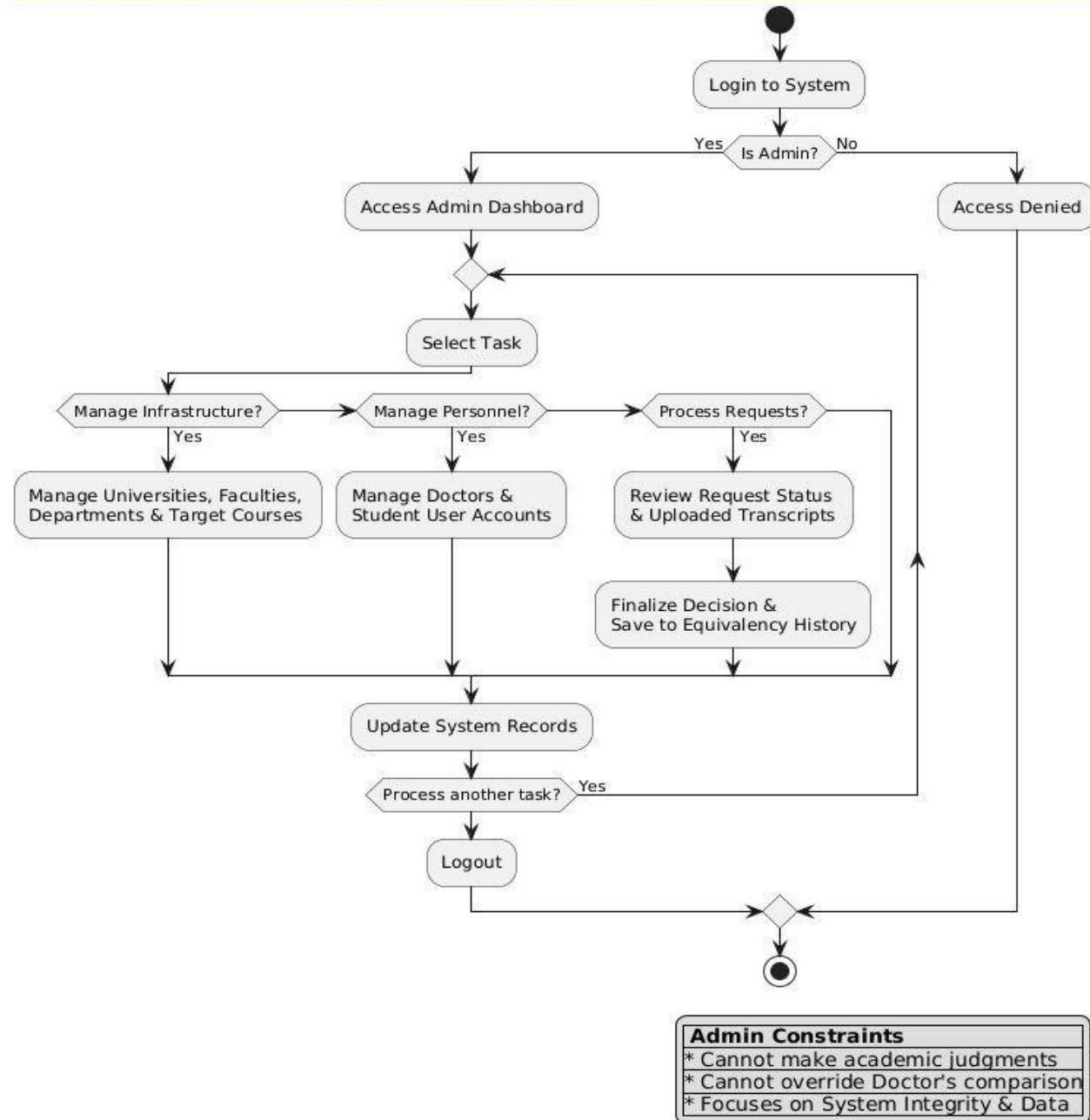
This syntax is deprecated, you must add <<#pink>> at the end of the line, after the ';'.

Activity Diagram:Admin

1:The admin is responsible for managing the system rather than making academic decisions.

2:After logging in, the admin can manage universities, courses, and user accounts,or monitor system requests and records.

3:The admin ensures data integrity and system organization,but cannot override the doctor's decisions or perform academic evaluations.



ERD:

The system database is divided into four main components:

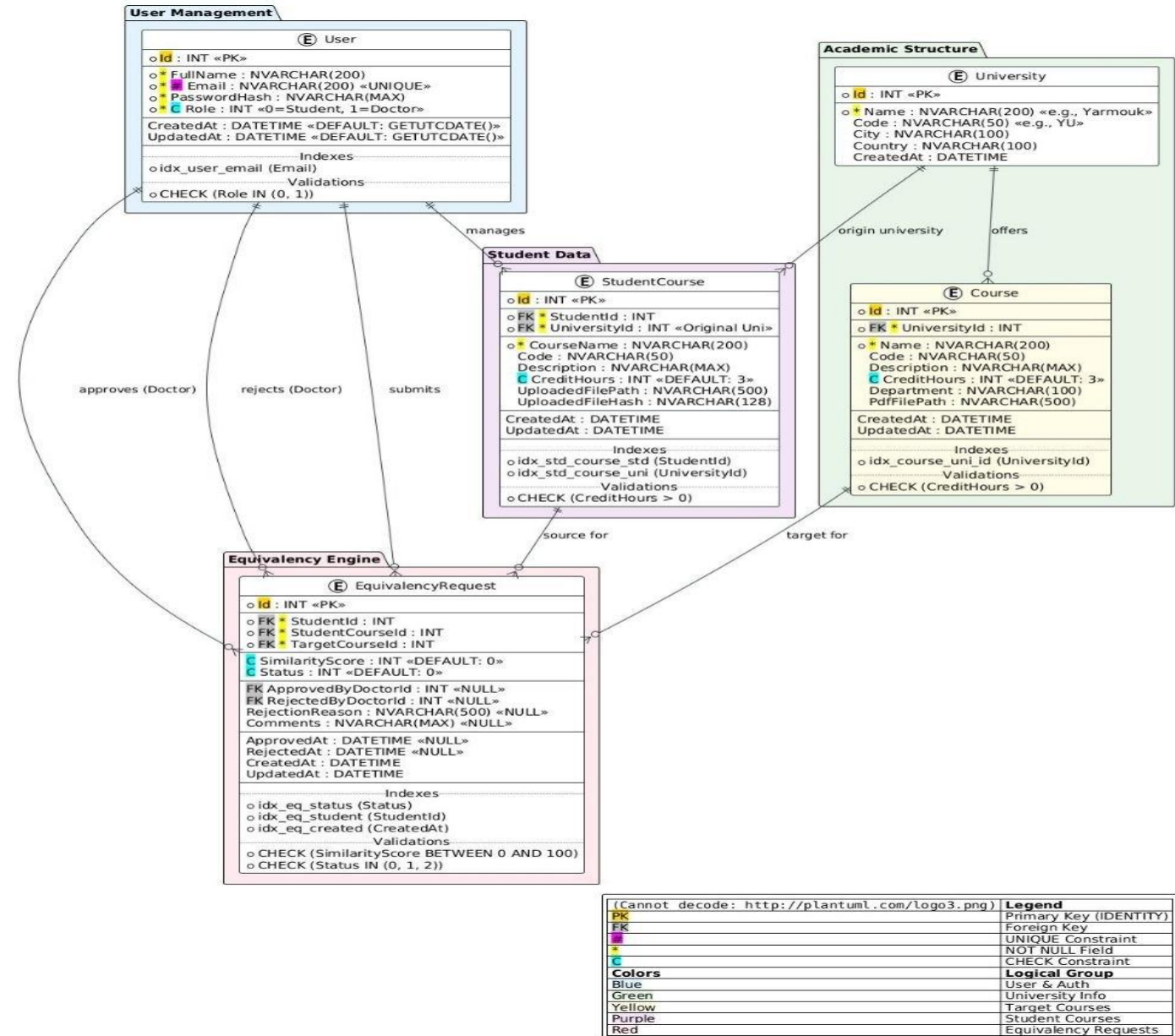
1: user management, academic structure, student data, and the equivalency engine.

2: Students upload their courses into the StudentCourse table, while official university courses are stored in the Course table.

3: When a student submits a request, it is stored in the EquivalencyRequest table, which links the student course with the target course.

4: The system calculates a similarity score using AI, and based on that, the request is either automatically approved or sent to a doctor for manual review.

5: All decisions, including approval or rejection, are stored along with timestamps and comments.



AI-Based Course Equivalency System - Database Schema

Relational Schema:

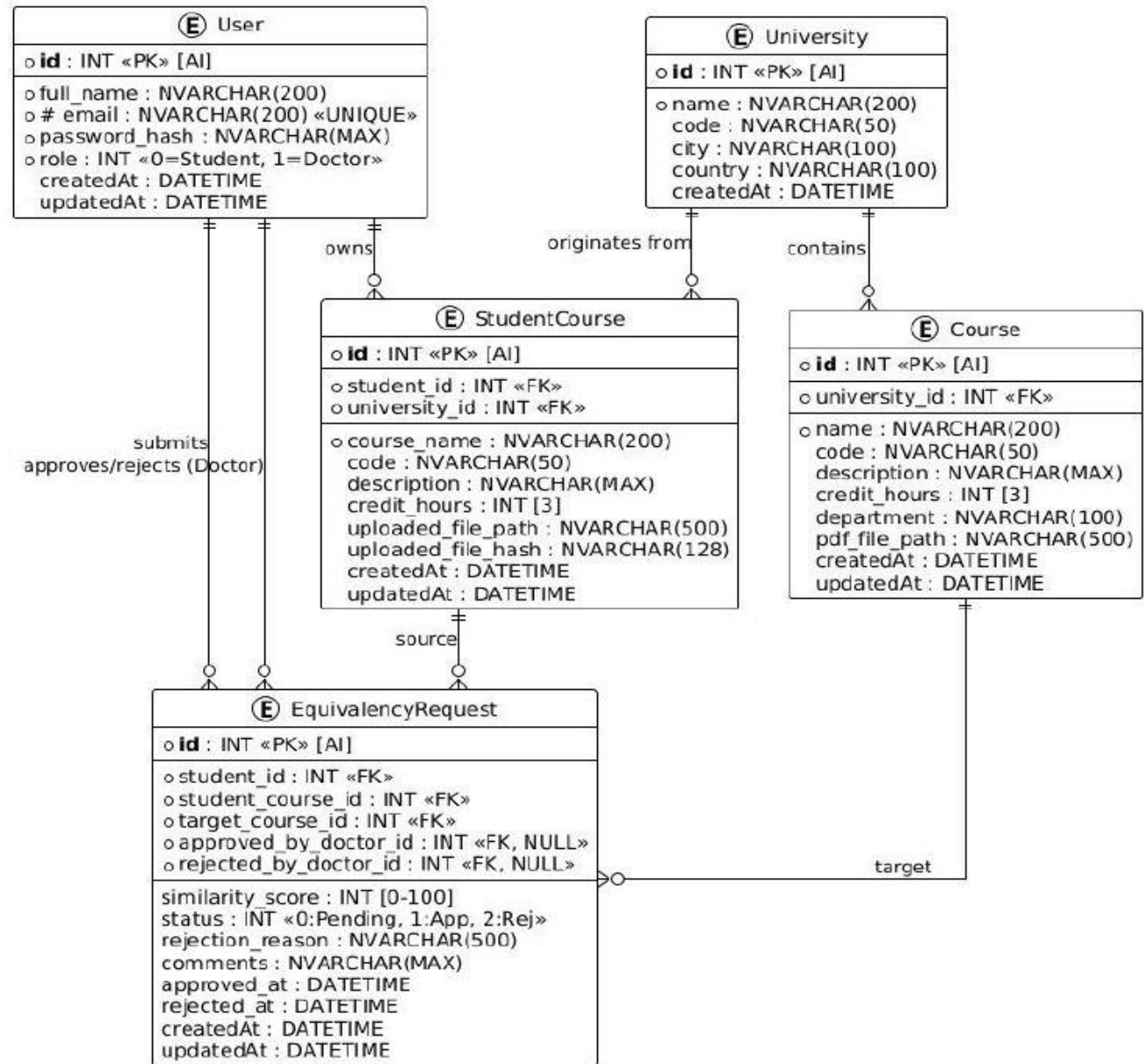
1:The system database consists of five main tables: User, University, Course, StudentCourse, and EquivalencyRequest.

2:Students upload their previous courses into the StudentCourse table,while official university courses are stored in the Course table.

3:When a student submits a request, it is saved in the EquivalencyRequest table,which links the student course with the target course.

4:The system calculates a similarity score using AI,and based on that, the request is either approved automatically,or reviewed by a doctor.

5:All decisions are stored with timestamps, comments, and status updates.



Use-Case Specifications

AI-Based Course Equivalency System

1 Upload Transcript 	
Actor	Student
Goal	Upload a transcript (PDF) to extract completed courses.
Preconditions	Student is logged in. File is in PDF format.
Main Flow	- Student selects a PDF file. - System uploads the file. - System confirms successful upload.
Alternative Flow	- Invalid file format → Show error. - Upload fails → Display retry message.
Postconditions	File is saved and linked to the student.

2 Extract Course Content 	
Actor	System
Goal	Extract text content from uploaded PDF files.
Preconditions	File exists in storage, File path is valid.
Main Flow	- System reads the PDF file. - Extracts text from all pages. - Stores the extracted text for processing.
Alternative Flow	File is unreadable or corrupted → Return empty text and log error
Postconditions	Course text content is available for comparison.

3 Create Equivalency Request 	
Actor	Student
Goal	Submit a request to check equivalency with a target course.
Preconditions	Student has uploaded at least one course.
Main Flow	- Student selects a previously uploaded course. - Student selects a target course. - System validates a request with status "Pending"
Alternative Flow	Course or target course not found → Show error message.
Postconditions	Equivalency request is stored in the database.

4 Compare Courses (AI Analysis) 	
Actor	System (AI Module)
Goal	Calculate similarity between two courses using AI embeddings.
Preconditions	Both course texts are available OpenAI API is configured and accessible.
Main Flow	- System sends course texts to OpenAI embeddings API, - Receives vector embeddings. - Calculates cosine similarity.
Alternative Flow	API request fails → Return similarity score as 0.
Postconditions	Similarity score is stored with the request.

5 Review Equivalency Request 	
Actor	Doctor
Goal	Review the request. AI similarity score, and decide to approve or
Preconditions	Doctor is logged in. Request exists with status "Pending".
Main Flow	- Doctor opens the request. - Reviews student info, course documents, and similarity score. - Approves or rejects the request.
Alternative Flow	(If rejected) adds reason or canimernity.
Postconditions	Decision is saved (approved or Rejected).

6 View Results and Status 	
Actor	Student
Goal	View the decsion and similarity score of submitted requests.
Preconditions	Student is logged in. At least one request exists.
Main Flow	- Student opens the results page. - System displays a list of requets - Student views status (Pending / Approved / Rejected)
Alternative Flow	No requests found → Show empty state message.
Postconditions	Student is informed about the final decision.

Conclusion:

- **This project presents an AI-based course equivalency system designed to address the inefficiencies of the traditional manual process. By automating key operations and incorporating intelligent similarity analysis, the system significantly reduces processing time and minimizes redundant work.**
- **The integration of AI techniques enhances the accuracy and consistency of equivalency decisions while supporting academic staff in making faster and more informed judgments. Furthermore, the system improves the overall user experience by providing students with a clear and efficient platform to submit and track their requests.**
- **By maintaining a structured history of past decisions, the system enables knowledge reuse and continuous improvement over time. In conclusion, this solution demonstrates the potential of combining automation and AI to modernize and optimize academic workflows.**